

Smart Environment Monitoring System Using Arduino UNO

Components Required

Arduino UNO

DHT22 Temperature & Humidity Sensor

MQ2 Gas Sensor

16×2 I2C LCD Display

Jumper Wires

USB Cable / Power Supply

Breadboard (Optional)

Pin Connections

DHT22 Sensor

DHT22 Pin Arduino UNO

VCC 5V

GND GND

DATA D7

MQ2 Gas Sensor

MQ2 Pin Arduino UNO

VCC 5V

MQ2 Pin Arduino UNO

GND GND

AO A0

LCD (I2C)

LCD Pin Arduino UNO

VCC 5V

GND GND

SDA A4

SCL A5

Working Principle

The Arduino UNO reads temperature and humidity values from the DHT22 sensor and gas concentration from the MQ2 sensor. These values are displayed on the 16×2 LCD.

- If the gas value is **below the threshold**, the LCD displays "**Gas: SAFE**".
- If the gas value is **above the threshold**, the LCD displays "**Gas: DANGER**".

This system helps in monitoring environmental conditions and detecting harmful gas leakage.

Project Link

<https://wokwi.com/projects/468283437244832769>